Correlated, nested subqueries

2026. February 22., Sunday 20:53

Correlated subqueries

Use values from the outer query to generate a result, re-run for every row in the created dataset
Dependent on the main query opposed to a simple subquery

A correlated example

```
SELECT
    s.stage,
    ROUND(s.avg_goals,2) AS avg_goal,
    (SELECT AVG(home_goal + away_goal)
     FROM match
     WHERE season = '2012/2013') AS overall_avg
FROM
    (SELECT
         stage,
         AVG(home_goal + away_goal) AS avg_goals
     FROM match
     WHERE season = '2012/2013'
     GROUP BY stage) AS s
WHERE s.avg_goals > (SELECT AVG(home_goal + away_goal)
                     FROM match AS m
                     WHERE s.stage > m.stage);
```

Nested subqueries
Can be correlated or uncorrelated-it can be referenced from an outer subquery or main query

...inside a subquery!

- *How does each month's total goals differ from the **average monthly total** of goals scored?*

```
SELECT
  EXTRACT(MONTH FROM date) AS month,
  SUM(m.home_goal + m.away_goal) AS total_goals,
  SUM(m.home_goal + m.away_goal) -
  (SELECT AVG(goals)
   FROM (SELECT
           EXTRACT(MONTH FROM date) AS month,
           SUM(home_goal + away_goal) AS goals
         FROM match
         GROUP BY month)) AS avg_diff
FROM match AS m
GROUP BY month;
```

Correlated nested subquery

- *What is the each country's average goals scored in the 2011/2012 season?*

```
SELECT
  c.name AS country,
  (SELECT AVG(home_goal + away_goal)
   FROM match AS m
   WHERE m.country_id = c.id -- Correlates with main query
         AND id IN (
             SELECT id -- Begin inner subquery
             FROM match
             WHERE season = '2011/2012')) AS avg_goals
FROM country AS c
GROUP BY country;
```